Panduru Sri Abish

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.49
Tirumala Junior College	Class XII (Board of Intermediate Education, AP)	2024	98.50%
Sri Chaitanya Techno School	Class X (Board of Secondary Education, AP)	2022	96.33%

SKILLS

Languages - Python, C,C++,HTML/CSS, JavaScript, SQL.

Frameworks - PyTorch, TensorFlow, NLTK, Scikit-learn, Numpy, Pandas, Matplotlib, Seaborn, OpenCV.

Tools : Git/Github, MySQL

RELEVANT COURSES

Mathematics: Linear Algebra, Calculus, Introduction to Statistics, Introduction to Probability, Differential Equations.

Computer Science: Introduction to Programming, Introduction to Artificial Intelligence, Data Structures and Applications.

PROJECTS

Credit Card Fraud Detection Pipeline | Github

- Built an end-to-end credit card fraud detection system, tackling the classic challenge of highly imbalanced data using SMOTE to balance the classes before training.
- Trained and compared five models -- from Logistic Regression to XGBoost -- to find the one that best separated fraud from genuine transactions.
- XGBoost came out on top with a ROC-AUC of ~0.99, and the whole pipeline was built as a modular system ready to move from ingestion to deployment.

PPE Safety Compliance Detector | Github

- Fine-tuned YOLOv8n on a 10-class construction-safety dataset (~2,800 images) using a two-stage training approach -- frozen backbone warm-up, then full fine-tuning.
- Achieved 76.4% mAP@0.5 at ~80 FPS, meeting real-time performance requirements for live video monitoring.
- Built custom logic to match detected PPE (hardhat, vest) to the correct person using bounding-box overlap, turning raw detections into per-worker compliance decisions.
- Deployed as a Streamlit app with configurable compliance rules and downloadable violation logs.

Domain-Specific RAG Q&A Assistant for arXiv ML Research *(in progress)* | Github

Python, FAISS, Sentence-Transformers, Streamlit

- Building a **retrieval-augmented generation** pipeline over a 199-paper **arXiv** corpus, covering ingestion, sentence-aware **chunking**, and embedding generation
- Implementing **dense vector retrieval** using **FAISS** and **Sentence-Transformers (MiniLM)** to surface relevant paper excerpts for a given query
- Integrating **LLM-based answer generation** with enforced source **citations**, deploying an interactive **Streamlit** interface for grounded, verifiable answers

Credit Card Customer Intelligence & Churn Analytics | Github

- Analyzed 10,127 credit card customer records to identify churn drivers, engineering features (utilization buckets, engagement score, inactivity rate) after cleaning leaked target-correlated columns from the raw dataset
- Built rule-based customer segmentation (Dormant, At Risk, High Value, Premium) using quantile-driven thresholds, uncovering a **64.5% churn rate in Dormant customers vs. 12% baseline** -- enabling targeted retention prioritization
- Wrote analytical SQL (window functions, CTEs, views) in SQLite to rank customers by spend, compute churn rates by inactivity level, and build a reusable retention-watchlist view for at-risk accounts
- Conducted EDA across demographics, spending behavior, and correlation analysis in Python (Pandas, Matplotlib, Seaborn), concluding churn is driven by customer activity/engagement rather than income or demographics

POSITIONS OF RESPONSIBILITY

Core Member at **Torque**(**The Automobile Engineering and Design Club, IIT Hyderabad**)

- Designed and developed the race track layout for **Remote Rumble**, an RC car racing event conducted during **Milan.**
- Collaborated with club members to organize technical events, workshops, and competitions for the student community.

Coordinator at **Prayas**(RDC),**IIT Hyderabad**

- Conducted Campus tours for School students, assisted in organizing outreach activities as part of the **Prayas** community engagement initiatives.
- Coordinated with volunteers to ensure the smooth execution of educational and community outreach events.

EXTRACURRICULAR

Conducted guided campus tours under **VISIT-IITH** for visitors at **IIT Hyderabad**, effectively communicating the institute's infrastructure, culture and student experience.

National Sports Organization (NSO) **Hockey Player** | IIT Hyderabad